

Smart Library Management System

WE_222_2.2



Agenda

1. Introduction
2. Methodology
3. Components
4. Conclusion



Introduction

The Student Book Borrowing System is designed to make the library process easier and more organized. It allows students to search for books, request to borrow them, and reserve books when they are not available.

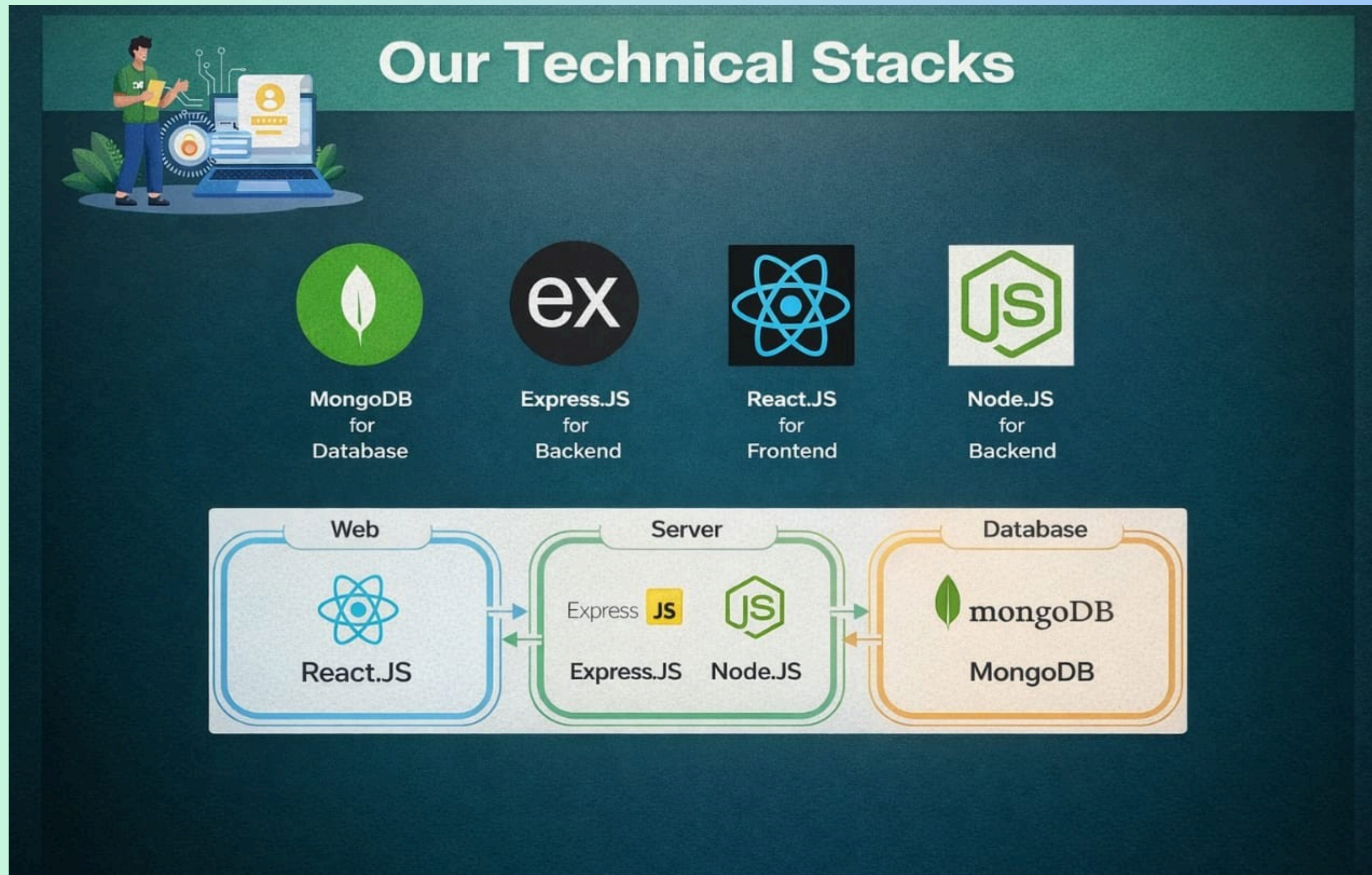
Why This System?



- Reduces manual library work
- Saves time for students and staff
- Provides real-time book availability
- Ensures fair reservation using queue system
- Improves overall efficiency of library management



Methodology



Components

1. Search Book & Borrow management
2. AI-Based Smart Assistant & Chatbot Module
3. Admin Book, Borrow & Inquiry Management Module
4. Library Space Management & E-Learning Module

Search Book & Borrow management

IT23202290
W.M.L.M. Premarathna



Main Functions

User Authentication - Secure student login using email and password.



Search & View Books - Search for book by title, author or category; check availability.



Borrow or Join Waiting List - Borrow up to 3 books at a time if available
OR
Join waiting list (FIFO method).



Request Management - View requests, cancel or edit before admin approval.



Admin Validation & Approval - Staff reviews and approves requests.



Notification & Completion - Student receives notification after approval.



Return & Fine Calculation - System calculates fine automatically if returned late.

Special Features

- Join Waiting List
- Inquiry management
- Fine calculation



AI-Based Smart Assistant & Chatbot Module

IT23208230
K.K.D Bandara



Main Functions

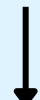
User Authentication – Secure member login.



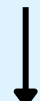
Data Collection & Analysis – System analyzes borrow history, return behavior, preferred categories, and due dates.



Personalized Recommendation Generation – Suggest books based on user interests and trending titles; exclude unavailable books.



Overdue Risk Evaluation – Calculate risk level (Low / Medium / High) using past late returns and remaining due days.



Automatic Fine Processing – If a book is returned late, system calculates fine and updates member records.



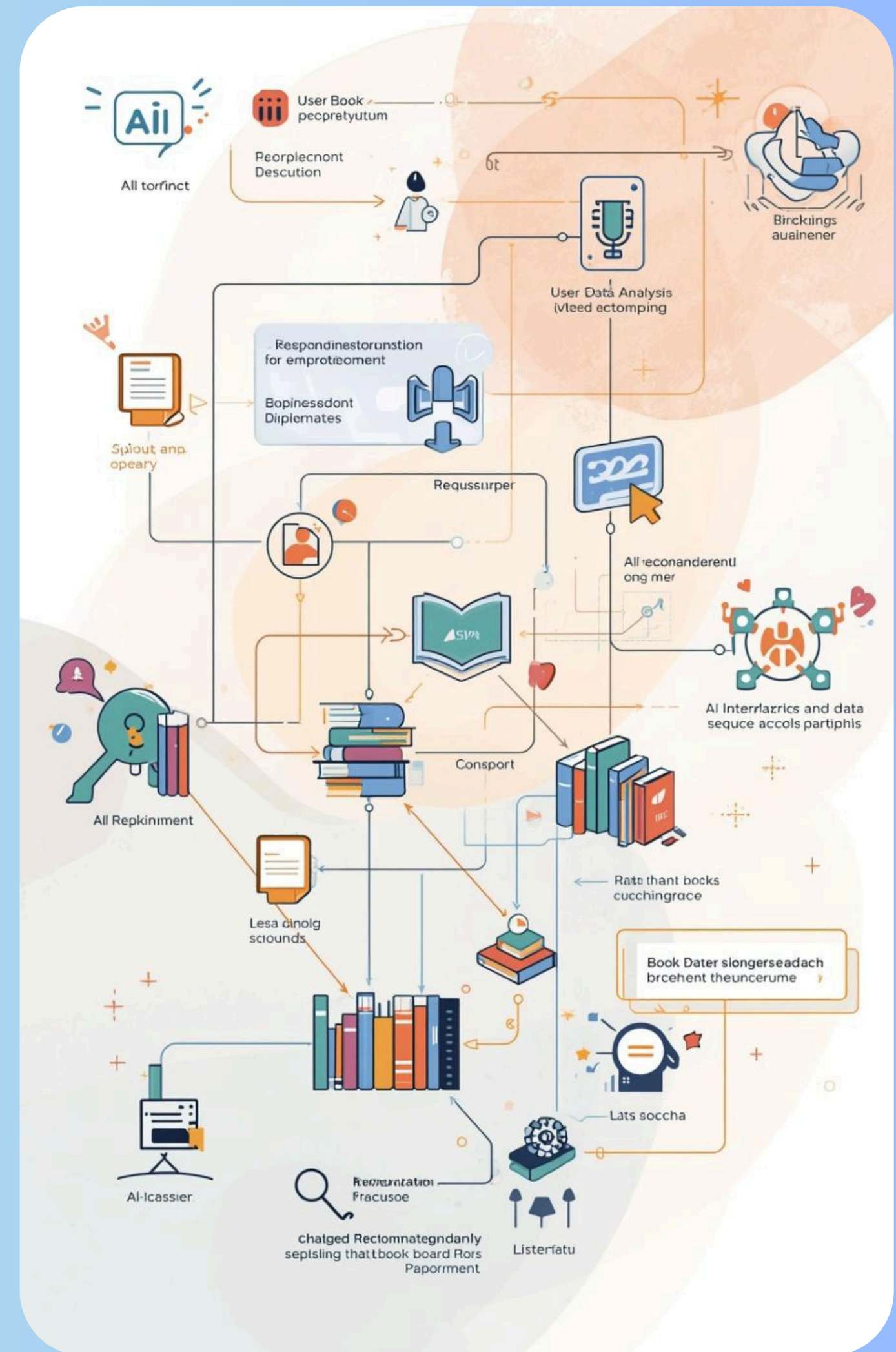
AI Chatbot Interaction – Users ask questions about book availability, borrowing rules, reservations, inquiries, or e-learning navigation.



Response, Notification & Record Storage – System provides instant responses and saves recommendations, risk data, fines, and chat logs.

Special Features

- Book recommendations
- Overdue Risk Prediction
- AI Chatbot



Admin Book, Borrow & Inquiry Management Module

IT23244320
K.G.A.I.Jayarathna



Main Functions

Admin Authentication – Secure admin login to access management functions.



Book Management – Add new books, update details, or remove damaged/obsolete books.



Borrow Request Monitoring – View student borrow requests and waiting lists.



Request Validation & Approval – Approve or reject borrow requests based on availability and rules.



Overdue Tracking & Report Generation – Monitor late returns and generate system reports.



Inquiry Management – View student inquiries, send replies, and update status (Pending / Answered / Closed).



System Update & Activity Logging – All actions are saved and recorded for tracking and auditing.

Special Features

- **Centralized Admin Dashboard**
- **Real-time Borrow Approval System**
- **Structured Inquiry Management**
- **Activity Logging & Report Generation**



Library Space Management & E-Learning Module

IT23218130
Manikkrama S.S.S.S



Main Functions

Admin Authentication – Secure login to manage spaces and learning resources.



Library Space Setup – Add reading rooms, seats, and define available time slots.



Space Reservation Process – Students check availability and reserve study spaces.



Reservation Validation & Monitoring – System prevents overlapping bookings and auto-releases unused reservations.



E-Learning Content Management – Admin uploads materials, creates categories, and updates digital resources.



Student Access to Learning Resources – Students view, download notes, and watch educational videos.



System Update & Usage Tracking – Store reservation data and generate usage reports.

Special Features

- **Smart Space Reservation System**
- **Integrated E-Learning Platform (New Feature)**
- **Role-Based Access Control**
- **Usage Monitoring & Reports**



Conclusion

The Smart Library Management System improves efficiency, accessibility, and overall user experience.

It streamlines library operations, supports student learning, and modernizes traditional library processes through smart and digital solutions.

This system helps create a more organized, efficient, and student-friendly library environment.

Thank You

